

LINDEN AGCM, AB, Canada

WMO: 710350

Lat: 51.6192N

Lon: 113.6559W

Elev: 914

StdP: 90.82

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -32.7	(c) -28.7	(d) -36.1	(e) 0.1	(f) -32.7	(g) -31.8	(h) 0.2	(i) -28.5	(j) 12.0	(k) -3.9	(l) 10.2	(m) -0.3	(n) 1.6	(o) 100	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.6	(c) 28.4	(d) 16.2	(e) 26.6	(f) 15.6	(g) 24.9	(h) 15.2	(i) 18.8	(j) 25.2	(k) 17.6	(l) 23.9	(m) 16.5	(n) 22.6	(o) 3.6	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.3	13.0	22.2	15.1	12.0	20.9	14.0	11.1	19.7	57.2	25.4	53.3	24.0	49.8	22.6	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	9.1	7.9	-37.9	32.4	3.0	2.1	-40.1	33.9	-41.9	35.2	-43.6	36.4	-45.8	37.9		
			-38.0	21.4	3.0	1.4	-40.2	22.4	-42.0	23.2	-43.6	24.0	-45.8	25.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-9.6	-10.6	-4.3	2.7	9.5	13.4	15.6	14.9	10.4	3.4	-4.6	-11.3	(6)	(7)
	DBStd	11.55	8.36	8.40	8.05	4.99	4.11	2.85	2.41	2.86	4.46	4.92	7.61	8.36	(7)	(8)
	HDD10.0	3272	608	577	442	221	60	6	0	2	47	210	439	661	(8)	(9)
	HDD18.3	5777	866	811	701	469	274	149	89	109	238	463	688	919	(9)	(10)
	CDD10.0	549	0	0	0	3	45	107	173	155	60	6	1	0	(10)	(11)
	CDD18.3	11	0	0	0	0	0	1	4	4	1	0	0	0	(11)	(12)
	CDH23.3	736	0	0	0	5	64	73	207	247	134	6	0	0	(12)	(13)
	CDH26.7	152	0	0	0	0	8	9	41	62	33	0	0	0	(13)	(14)
(14) Wind	WSAvg	3.5	3.1	3.6	3.2	3.6	3.9	4.1	3.2	2.9	3.2	4.1	3.4	3.3	(14)	(15)
(15) Precipitation	PrecAvg	516	19	16	25	38	68	99	73	63	45	26	24	20	(15)	(16)
	PrecMax	678	37	32	54	87	110	254	160	114	110	51	60	44	(16)	(17)
	PrecMin	327	4	3	8	15	24	44	28	12	7	8	5	6	(17)	(18)
	PrecStd	68	8	7	10	17	25	39	31	24	24	11	12	10	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.4	8.7	15.5	23.3	27.8	28.0	30.0	31.3	30.2	23.5	17.2	8.7	(19)	(20)
		MCWB	3.8	3.7	7.1	9.3	12.4	16.1	18.1	16.7	14.5	10.9	8.0	3.0	(20)	(21)
	2%	DB	6.1	5.4	11.9	19.1	25.2	25.4	27.7	28.2	27.0	19.0	12.0	5.1	(21)	(22)
		MCWB	2.5	1.9	4.8	8.4	11.6	14.8	18.1	16.6	13.7	9.2	5.7	1.7	(22)	(23)
	5%	DB	4.5	3.3	9.0	15.5	22.5	23.4	25.6	26.1	24.5	16.2	8.4	3.4	(23)	(24)
		MCWB	1.6	0.4	3.5	6.4	10.8	13.9	17.7	16.4	13.0	8.0	3.5	0.6	(24)	(25)
	10%	DB	2.3	1.1	5.9	12.4	20.0	21.6	23.8	23.9	21.1	13.2	5.6	0.6	(25)	(26)
		MCWB	0.0	-0.8	2.1	4.9	10.0	13.0	16.6	15.8	11.7	6.7	1.9	-1.4	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	4.1	7.5	10.5	13.3	18.3	21.1	19.5	15.9	12.1	8.4	3.6	(27)	(28)
		MCDB	8.5	8.0	14.4	21.9	24.3	25.2	26.3	26.1	26.7	21.5	15.8	8.0	(28)	(29)
	2%	WB	2.7	2.1	5.5	8.5	12.2	16.4	19.4	18.3	14.7	10.1	5.9	2.0	(29)	(30)
		MCDB	6.0	5.1	11.0	18.2	22.9	23.0	25.6	25.0	24.5	17.3	11.4	5.0	(30)	(31)
	5%	WB	1.6	0.7	3.7	6.9	11.4	15.0	18.1	17.3	13.4	8.6	3.8	0.5	(31)	(32)
		MCDB	4.4	3.0	8.4	14.0	20.9	20.9	24.1	23.9	22.4	15.6	7.7	3.1	(32)	(33)
	10%	WB	0.0	-0.8	2.2	5.3	10.4	13.8	17.1	16.1	12.3	7.1	2.2	-1.5	(33)	(34)
		MCDB	2.3	1.1	5.7	11.4	18.6	19.9	23.0	22.6	20.9	12.7	5.2	0.7	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.4	12.1	13.6	15.0	13.4	14.6	15.8	16.2	15.0	12.7	11.7	(35)	(36)
		MCDBR	12.3	14.1	16.2	21.1	21.0	18.0	18.9	21.2	23.3	20.1	16.6	13.8	(36)	(37)
		MCWBR	9.1	10.7	9.8	11.0	9.1	8.5	10.6	10.7	10.8	11.1	10.6	10.4	(37)	(38)
	5% WB	MCDBR	11.8	13.2	15.2	19.8	18.1	15.3	17.2	18.1	20.9	19.3	15.9	13.6	(38)	(39)
		MCWBR	9.1	10.2	9.5	10.5	8.0	8.3	10.3	10.1	9.9	10.7	10.5	10.5	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.258	0.272	0.283	0.331	0.342	0.344	0.344	0.359	0.313	0.281	0.253	0.239	(40)	(41)
	taud		2.359	2.407	2.459	2.347	2.374	2.405	2.409	2.370	2.458	2.531	2.479	2.362	(41)	(42)
	Ebn at Noon		775	870	927	908	911	910	903	869	879	852	790	742	(42)	(43)
	Edh at Noon		59	76	88	111	114	111	109	108	87	66	52	50	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		1.10	2.11	3.48	4.71	5.68	6.06	6.49	5.27	3.71	2.25	1.20	0.86	(44)	(45)
	RadStd		0.08	0.17	0.28	0.37	0.39	0.38	0.34	0.37	0.40	0.26	0.10	0.08	(45)	(46)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page